

Big Data Management – Assignment 01

Student Details

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1. Introduction

This assignment demonstrates the implementation of Hadoop MapReduce to analyze CRAN package download logs. The dataset contains information about package downloads including date, package name, country, and operating system. MapReduce jobs were implemented to answer analytical queries about package usage.

2. Environment Setup

Operating System: Ubuntu (Virtual Machine)

Framework: Apache Hadoop

Programming Language: Java

Terminal screenshot showing Hadoop services running (jps):

```
piyush — piyush3191958@UbuntuServer24: ~ — ssh piyush3191958@1...  
[piyush3191958@UbuntuServer24:~$ hadoop version  
Hadoop 3.4.3  
Source code repository https://github.com/apache/hadoop.git -r 9d50c6884666e794e  
45102260a4017bb31802e1b  
Compiled by stevel on 2026-02-12T15:31Z  
Compiled on platform linux-aarch_64  
Compiled with protoc 3.23.4  
From source with checksum 2331238c4c2929e66645316a32a8613  
This command was run using /usr/local/hadoop/share/hadoop/common/hadoop-common-3  
.4.3.jar  
piyush3191958@UbuntuServer24:~$ start-dfs.sh  
[start-yarn.sh  
Starting namenodes on [localhost]  
Starting datanodes  
Starting secondary namenodes [UbuntuServer24]  
Starting resourcemanager  
Starting nodemanagers  
piyush3191958@UbuntuServer24:~$ jps  
2561 ResourceManager  
2772 Jps  
2694 NodeManager  
2374 SecondaryNameNode  
2089 NameNode  
piyush3191958@UbuntuServer24:~$ █
```

3. Dataset Preparation

Dataset Source: CRAN logs

Steps performed:

- Download dataset
- Extract CSV file
- Upload dataset to HDFS

```
piyush — piyush3191958@UbuntuServer24: ~ — ssh piyush3191958@1...

2026-03-11 12:04:40 (4.29 MB/s) - '2026-03-02.csv.gz' saved [115038768/115038768]

[piyush3191958@UbuntuServer24:~$ ls
2026-03-02.csv.gz      hadoop-3.3.6.tar.gz      test_db      wget-log.1
2026-03-02.csv.gz.1   hadoop-3.4.3-aarch64.tar.gz  wget-log     word-count-example
[piyush3191958@UbuntuServer24:~$ gunzip 2026-03-02.csv.gz
[piyush3191958@UbuntuServer24:~$ head 2026-03-02.csv
"date","time","size","r_version","r_arch","r_os","package","version","country","
ip_id"
"2026-03-02","17:44:15",48514,"4.5.2","x86_64","mingw32","sys","3.4.3","FR",1
"2026-03-02","17:44:16",166987,"4.5.2","x86_64","mingw32","gridBase","0.4-7","FR
",1
"2026-03-02","17:44:16",253794,"4.5.2","x86_64","mingw32","rlist","0.4.6.2","FR
",1
"2026-03-02","17:44:16",3367291,"4.5.2","x86_64","mingw32","ragg","1.5.0","FR",1
"2026-03-02","17:44:18",2282322,"4.5.2","x86_64","mingw32","ggridges","0.5.7",NA
,2
"2026-03-02","17:44:19",1089263,NA,NA,NA,"pan","1.9",NA,2
"2026-03-02","17:44:18",365,"4.5.2","x86_64","linux-gnu","plyr","1.8.9","BR",3
"2026-03-02","17:44:14",77992,NA,NA,NA,"DiallelAnalysisR","0.5.0","US",4
"2026-03-02","17:43:46",227085,"4.5.2","x86_64","mingw32","digest","0.6.39","US"
,5
piyush3191958@UbuntuServer24:~$ █
```

```
piyush — piyush3191958@UbuntuServer24: ~ — ssh piyush3191958@1...

tcp      0      0 0.0.0.0:9867      0.0.0.0:*        LISTEN
9156/java
tcp      0      0 0.0.0.0:9866      0.0.0.0:*        LISTEN
9156/java
tcp      0      0 0.0.0.0:9868      0.0.0.0:*        LISTEN
9320/java
tcp      0      0 0.0.0.0:9870      0.0.0.0:*        LISTEN
9027/java
tcp      0      0 0 127.0.0.1:8020   0.0.0.0:*        LISTEN
9027/java
tcp      0      0 0.0.0.0:13562     0.0.0.0:*        LISTEN
9759/java
tcp      0      0 0.0.0.0:35985     0.0.0.0:*        LISTEN
9759/java
[piyush3191958@UbuntuServer24:~$ hdfs dfs -mkdir /cran
[piyush3191958@UbuntuServer24:~$ hdfs dfs -ls /cran
[piyush3191958@UbuntuServer24:~$ hdfs dfs -mkdir /cran
mkdir: `/cran': File exists
[piyush3191958@UbuntuServer24:~$ hdfs dfs -put 2026-03-02.csv /cran
[piyush3191958@UbuntuServer24:~$ hdfs dfs -ls /cran
Found 1 items
-rw-r--r--  1 piyush3191958 supergroup 661929041 2026-03-11 12:40 /cran/2026-0
3-02.csv
piyush3191958@UbuntuServer24:~$ █
```

4. MapReduce Implementation

Each query was implemented using Mapper, Reducer, and Driver classes in Java.

Query 1 – Number of downloads of ggplot2

Description: Count how many times the package ggplot2 was downloaded.

```
piyush — piyush3191958@UbuntuServer24: ~/assignment1/Q1 — ssh piyush3191958@192.168....

WRONG_MAP=0
WRONG_REDUCE=0
File Output Format Counters
  Bytes Written=14
2026-03-13 19:53:33,886 INFO mapred.LocalJobRunner: Finishing task: attempt_local1977907263_0001_r_000_000_0
2026-03-13 19:53:33,886 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-03-13 19:53:33,959 INFO mapreduce.Job: map 100% reduce 100%
2026-03-13 19:53:33,959 INFO mapreduce.Job: Job job_local1977907263_0001 completed successfully
2026-03-13 19:53:33,967 INFO mapreduce.Job: Counters: 36
File System Counters
  FILE: Number of bytes read=2122110
  FILE: Number of bytes written=9181184
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=2666109090
  HDFS: Number of bytes written=14
  HDFS: Number of read operations=63
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=8
  HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=8339509
  Map output records=74454
  Map output bytes=893448
  Map output materialized bytes=1042386
  Input split bytes=530
  Combine input records=0
  Combine output records=0
  Reduce input groups=1
  Reduce shuffle bytes=1042386
  Reduce input records=74454
  Reduce output records=1
  Spilled Records=148908
  Shuffled Maps =5
  Failed Shuffles=0
  Merged Map outputs=5
  GC time elapsed (ms)=56
  Total committed heap usage (bytes)=2228224000
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=661945425
File Output Format Counters
  Bytes Written=14
[piyush3191958@UbuntuServer24:~/assignment1/Q1$ hdfs dfs -cat /cran/output_q1/part-r-00000
2026-03-13 19:54:27,768 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your plat
form... using builtin-java classes where applicable
ggplot2 74454
piyush3191958@UbuntuServer24:~/assignment1/Q1$ █
```

Query 2 – Downloads per Country

Description: Count downloads grouped by country.

```

piyush — piyush3191958@UbuntuServer24: ~/assignment1/Q2 — ssh piyush3191958@192.168.56...

2026-03-13 22:48:49,299 INFO mapreduce.Job: Job job_local984798726_0002 running in uber mode : false
2026-03-13 22:48:49,300 INFO mapreduce.Job: map 100% reduce 100%
2026-03-13 22:48:49,300 INFO mapreduce.Job: Job job_local984798726_0002 completed successfully
2026-03-13 22:48:49,309 INFO mapreduce.Job: Counters: 36
  File System Counters
    FILE: Number of bytes read=1492050
    FILE: Number of bytes written=4201955
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=1324035600
    HDFS: Number of bytes written=144765
    HDFS: Number of read operations=51
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=14
    HDFS: Number of bytes read erasure-coded=0
  Map-Reduce Framework
    Map input records=5990
    Map output records=5990
    Map output bytes=120295
    Map output materialized bytes=132281
    Input split bytes=112
    Combine input records=0
    Combine output records=0
    Reduce input groups=5990
    Reduce shuffle bytes=132281
    Reduce input records=5990
    Reduce output records=1
    Spilled Records=11980
    Shuffled Maps =1
    Failed Shuffles=0
    Merged Map outputs=1
    GC time elapsed (ms)=2
    Total committed heap usage (bytes)=801112064
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=72375
  File Output Format Counters
    Bytes Written=15
piyush3191958@UbuntuServer24:~/assignment1/Q2$ hdfs dfs -ls /cran/q2_output
2026-03-13 22:48:59,092 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your plat
form... using builtin-java classes where applicable
Found 2 items
-rw-r--r--  3 piyush3191958 supergroup          0 2026-03-13 22:48 /cran/q2_output/_SUCCESS
-rw-r--r--  3 piyush3191958 supergroup        15 2026-03-13 22:48 /cran/q2_output/part-r-00000
piyush3191958@UbuntuServer24:~/assignment1/Q2$ hdfs dfs -cat /cran/q2_output/part-r-00000 | head
2026-03-13 22:49:06,460 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your plat
form... using builtin-java classes where applicable
"1.2.0" 186465
piyush3191958@UbuntuServer24:~/assignment1/Q2$ █

```

Query 3 – Top 10 Most Downloaded Packages

Description: Identify the most popular packages based on download counts.

```

piyush — piyush3191958@UbuntuServer24: ~/assignment1/Q3 — ssh piyush3191958@192.168.56...

HDFS: Number of bytes read=2666109090
HDFS: Number of bytes written=353427
HDFS: Number of read operations=63
HDFS: Number of large read operations=0
HDFS: Number of write operations=8
HDFS: Number of bytes read erasure-coded=0
Map-Reduce Framework
  Map input records=8339509
  Map output records=8339509
  Map output bytes=114880521
  Map output materialized bytes=1663471
  Input split bytes=530
  Combine input records=8339509
  Combine output records=97654
  Reduce input groups=25947
  Reduce shuffle bytes=1663471
  Reduce input records=97654
  Reduce output records=25947
  Spilled Records=195308
  Shuffled Maps =5
  Failed Shuffles=0
  Merged Map outputs=5
  GC time elapsed (ms)=60
  Total committed heap usage (bytes)=2026897408
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=661945425
File Output Format Counters
  Bytes Written=353427
piyush3191958@UbuntuServer24:~/assignment1/Q3$ hdfs dfs -ls /cran/q3_output
2026-03-13 22:54:15,996 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your plat
form... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 3 piyush3191958 supergroup 0 2026-03-13 22:53 /cran/q3_output/_SUCCESS
-rw-r--r-- 3 piyush3191958 supergroup 353427 2026-03-13 22:53 /cran/q3_output/part-r-00000
piyush3191958@UbuntuServer24:~/assignment1/Q3$ hdfs dfs -cat /cran/q3_output/part-r-00000 | head
2026-03-13 22:54:24,231 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your plat
form... using builtin-java classes where applicable
"A3" 10
"AATtools" 11
"ABACUS" 8
"ABC.RAP" 9
"ABCDscores" 9
"ABCanalysis" 57
"ABCOptim" 58
"ABHgenotypeR" 8
"ABM" 8
"ABPS" 9
cat: Unable to write to output stream.
piyush3191958@UbuntuServer24:~/assignment1/Q3$ █

```

Query 4 – Most Popular Package in Ireland

Description: Filter downloads from Ireland (IE) and count packages.


```
piyush — piyush3191958@UbuntuServer24: ~/assignment1/Q4 — ssh piyush3191958@192.168....

WRONG_MAP=0
WRONG_REDUCE=0
File Output Format Counters
  Bytes Written=17
2026-03-14 11:07:40,180 INFO mapred.LocalJobRunner: Finishing task: attempt_local1867445973_0001_r_0
00000_0
2026-03-14 11:07:40,181 INFO mapred.LocalJobRunner: reduce task executor complete.
2026-03-14 11:07:41,138 INFO mapreduce.Job: map 100% reduce 100%
2026-03-14 11:07:41,139 INFO mapreduce.Job: Job job_local1867445973_0001 completed successfully
2026-03-14 11:07:41,159 INFO mapreduce.Job: Counters: 36
  File System Counters
    FILE: Number of bytes read=42926
    FILE: Number of bytes written=3935235
    FILE: Number of read operations=0
    FILE: Number of large read operations=0
    FILE: Number of write operations=0
    HDFS: Number of bytes read=2666109090
    HDFS: Number of bytes written=17
    HDFS: Number of read operations=63
    HDFS: Number of large read operations=0
    HDFS: Number of write operations=8
    HDFS: Number of bytes read erasure-coded=0
  Map-Reduce Framework
    Map input records=8339509
    Map output records=29155
    Map output bytes=437325
    Map output materialized bytes=115
    Input split bytes=530
    Combine input records=29155
    Combine output records=5
    Reduce input groups=1
    Reduce shuffle bytes=115
    Reduce input records=5
    Reduce output records=1
    Spilled Records=10
    Shuffled Maps =5
    Failed Shuffles=0
    Merged Map outputs=5
    GC time elapsed (ms)=65
    Total committed heap usage (bytes)=2228224000
  Shuffle Errors
    BAD_ID=0
    CONNECTION=0
    IO_ERROR=0
    WRONG_LENGTH=0
    WRONG_MAP=0
    WRONG_REDUCE=0
  File Input Format Counters
    Bytes Read=661945425
  File Output Format Counters
    Bytes Written=17
[piyush3191958@UbuntuServer24:~/assignment1/Q4$ hdfs dfs -cat /cran/q4_output/part-r-00000 | head ]
2026-03-14 11:07:59,972 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your pl
atform... using builtin-java classes where applicable
2026-03-02      29155
piyush3191958@UbuntuServer24:~/assignment1/Q4$ █
```

Query 5 – Most Popular Operating System

Description: Count downloads grouped by operating system.

```
piyush — piyush3191958@UbuntuServer24: ~/assignment1/Q5 — ssh piyush3191958@192.168....

Map output bytes=64479212
Map output materialized bytes=1821
Input split bytes=530
Combine input records=5159279
Combine output records=99
Reduce input groups=24
Reduce shuffle bytes=1821
Reduce input records=99
Reduce output records=24
Spilled Records=198
Shuffled Maps =5
Failed Shuffles=0
Merged Map outputs=5
GC time elapsed (ms)=55
Total committed heap usage (bytes)=2146435072

Shuffle
Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=661945425
File Output Format Counters
  Bytes Written=387

[piyush3191958@UbuntuServer24:~/assignment1/Q5$ hdfs dfs -ls /cran/q5_output
2026-03-14 11:18:44,158 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your pl
atform... using builtin-java classes where applicable
Found 2 items
-rw-r--r-- 3 piyush3191958 supergroup 0 2026-03-14 11:18 /cran/q5_output/_SUCCESS
-rw-r--r-- 3 piyush3191958 supergroup 387 2026-03-14 11:18 /cran/q5_output/part-r-00000
piyush3191958@UbuntuServer24:~/assignment1/Q5$ hdfs dfs -cat /cran/q5_output/part-r-00000 | head -20
2026-03-14 11:18:52,995 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your pl
atform... using builtin-java classes where applicable
(4
darwin13.4.0 744
darwin15.6.0 2327
darwin17.0 36926
darwin20 986812
darwin20.0.0 323
darwin22.4.0 5
darwin22.6.0 1629
darwin23.0.0 5
darwin23.2.0 9
darwin23.4.0 366
darwin23.6.0 1067
darwin24.0.0 4
darwin24.2.0 18
darwin24.4.0 926
darwin24.6.0 3168
darwin25.0.0 5042
darwin25.1.0 5
darwin25.2.0 2556
darwin25.3.0 4537
piyush3191958@UbuntuServer24:~/assignment1/Q5$
```

5. Results Summary

Briefly summarize the findings from the MapReduce analysis. Example:

- ggplot2 had significant downloads on the selected date.
- Certain countries generated the highest traffic.
- Some packages appeared frequently among the top downloads.

6. Conclusion

The Hadoop MapReduce framework successfully processed the CRAN dataset and provided insights into package popularity, geographic distribution, and operating system usage. This assignment demonstrates how large-scale datasets can be analyzed using distributed computing.